BITS F464: Machine Learning

Assignment 1: Flat vs. Hierarchical Classification From Scratch
Semester I, Academic Year 2026-27 | Student ID: 2024ADPS0196H

Results Matrix

Dataset: Anuran Calls (MFCCs), 7,195 records, 22 MFCC features. Stratified 80/20 split on Species (seed 42): 5,756 training / 1,439 test rows. Flat models predict Species directly; hierarchical models route Family first, then Species. Accuracies are percentages; 100.00 means every training row correct.

Method	Criterion	Overall accuracy (Species)		Family acc. (Test)
		Train	Test	
1. Single Decision Tree	Gini	100.00	94.93	96.25
	Entropy	100.00	95.41	96.80
2a. Bagging	Gini	99.97	96.66	97.50
	Entropy	99.98	96.87	97.50
2b. AdaBoost	Gini	97.52	95.34	96.73
	Entropy	96.35	94.16	95.00
3. Random Forest	Gini	99.98	97.43	97.85
	Entropy	100.00	97.43	97.85
4. Hierarchical Decision Tree	Gini	100.00	93.82	Router: 95.07
	Entropy	100.00	94.65	Router: 95.62
5. Gaussian Naive Bayes	-	91.47	90.34	91.94

† For flat models, Family acc. is the derived family accuracy: each predicted species is mapped to its family using the species-to-family mapping read from the training data. For the hierarchical rows the shaded Family acc. cell contains the router accuracy, while the Overall accuracy columns contain the final species accuracy.

Table 3: Supplementary Hierarchical Failure Analysis (Test Set, %)

Criterion	Router acc.	Oracle acc.	(A) Router right, expert wrong	(B) Router wrong
Gini	95.07	98.26	1.25	4.93
Entropy	95.62	98.33	0.97	4.38

A_overall = A_router - pA and pB = 1 - A_router hold exactly at full precision for both criteria. Oracle accuracy uses the expert of the true family for every test row.

Fixed Settings

Single tree: unconstrained. Bagging: T = 25 bootstrap trees, all 22 features, majority vote. AdaBoost: multi-class SAMME, T = 50, depth-3 weak learners. Random forest: T = 50, m = floor(sqrt(22)) = 4 features per split, majority vote. Hierarchical: one unconstrained router and one unconstrained expert per family, same criterion. Gaussian NB: variance smoothing 1e-9 * max feature variance. All randomness comes from np.random.RandomState(42), one fresh generator per model; the split has its own seed-42 generator.

Generated from the submitted implementations; the five part scripts reproduce these numbers with one command each (see README.txt).